

Xin Yan

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Education

Sun Yat-sen University, Master of Science in Psychology Sept. 2024 – June 2027 (Expected)

- Major GPA: 4.0 / 5.0
- Selected Coursework: Research Methods in Psychology (92/100); Analysis of fMRI Data (93/100)

Sun Yat-sen University, Bachelor of Science in Psychology Sept. 2020 – June 2024

- Major GPA: 4.0 / 5.0
- Selected Coursework: Experimental Psychology (94/100); Fundamentals of MATLAB Programming (92/100)

Publications

Yan, X.*, Wang, W.*, Ye, S., & Ding, X. (2025). Eyes on the prize: Reward outranks punishment in working memory resources allocation. *Journal of Experimental Psychology: Human Perception and Performance*, 51(11), 1533 – 1546. <https://doi.org/10.1037/xhp0001363>

Yan, X.*, Zhu, B.*, & Mo, C. (2025). Effects of feature-based attention on numerosity perception. *Perception*, 54(5), 362-374. <https://doi.org/10.1177/03010066251326828>

Su, Z., Duan, Z., **Yan, X.**, Lin, Z., & Ding, X. (2025). Event boundaries: Costs and benefits for memory. *Cognition*, 264, 106239. <https://doi.org/10.1016/j.cognition.2025.106239>

Wang, W.*, **Yan, X.***, He, X.*, & Qian, J. (2024). Evidence for the beneficial effect of reward on working memory: A meta-analytic study. *Journal of Intelligence*, 12(9), 88. <https://doi.org/10.3390/jintelligence12090088>

Chen, W., Ye, S., **Yan, X.**, & Ding, X. (2024). The combination operation of grouping and ensemble coding for structured biological motion crowds in working memory. *Cognitive Research: Principles and Implications*, 9(1), 45. <https://doi.org/10.1186/s41235-024-00574-6>

(*Co-first authorship)

Research Experience

Neural Correlates of Consciousness: A Novel No-Report Paradigm 2024 – Present

- Independently designed an innovative no-report paradigm, incorporating high-load distractor tasks to isolate neural substrates of phenomenal consciousness from

post-perceptual processes.

- Led data acquisition for 20 participants, integrating simultaneous fMRI and eye-tracking recordings to objectively monitor implicit stimulus processing.
- Preprocessed fMRI data and conducted ROI-based and searchlight MVPA analyses using CoSMoMVPA.
- Prepared a full manuscript draft showing that conscious perceptual contents were reliably decoded from posterior cortical regions under high attentional load, whereas frontal decoding was mainly observed during passive viewing.

Hierarchical Representation and Flexible Retrieval of Cognitive Maps 2024 – Present

- Collaborated on the experimental design and data collection for 30 participants to investigate how task demands modulate the hierarchical retrieval of cognitive maps.
- Performed RSA to decode level-specific information readout, uncovering how the OFC encodes coarse global schemas while the hippocampus retrieves fine-grained sub-maps under high-precision demands.

Honors and Awards

2026	ECVP 2026 Travel Award (£500) European Conference on Visual Perception
2025	National Scholarship The Ministry of Education of the People's Republic of China
2025	First Prize Scholarship of Outstanding Students Sun Yat-sen University
2024	Outstanding Graduate Sun Yat-sen University
2024	First Prize in Psychological Experiment Design National Undergraduate Innovation & Entrepreneurship Forum

Professional Skills

Statistical Analysis:	Linear models, PCA, MVPA, RSA, neural networks
Methodologies:	Psychophysics, fMRI, Eye-tracking, simultaneous eye-tracking + fMRI, EEG
Programming skills:	MATLAB, Python, R
Languages:	English (proficient, TOEFL 107), Mandarin (native)

Teaching Experience

2025	Teaching Assistant, Academic writing (Xiaowei Ding) Sun Yat-sen University
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